PROJECT DOCUMENTATION

FORGE

Problem Statement Document

AI-Powered Fitness & Wellness Plan Builder

Version	Authors	Status	Type
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1. Executive Summary

FORGE is an AI-driven, multi-agent system purpose-built to solve one of the most persistent gaps in the consumer health and wellness market: the inability of generic, one-size-fits-all fitness content to account for individual variation. Through a coordinated pipeline of four specialized agents — each powered by Google Gemini and grounded in real-time web research — FORGE generates comprehensive, evidence-based, personalized 4-week fitness and nutrition plans that rival the output of professional personal trainers and nutritionists.

The system is deployed as a live web application on Railway and designed with an agentic architecture that decomposes the complex plan-generation task into discrete, verifiable steps — each contributing a specialized layer of intelligence and culminating in an LLM-as-Judge evaluation of scientific validity.

2. Problem Context & Background

2.1 The Fitness Information Problem

The global fitness industry generates billions of dollars in content annually — workout videos, nutrition blogs, app-based programs — yet the vast majority of this content is completely generic. A beginner with a knee injury who follows the same plan as an advanced athlete is not just uninspired; they are at risk of harm. A vegan individual following a protein plan built around animal sources will be left with dangerously inadequate macronutrient targets.

This mismatch between available content and individual need is not a minor inconvenience. It is a structural failure that causes people to abandon their fitness goals, sustain preventable injuries, or spend money on professional services that most cannot afford.

2.2 The Accessibility Problem

Personalized guidance from a qualified personal trainer or registered nutritionist remains a luxury product. Hourly rates are prohibitive for the majority of fitness seekers globally, and ongoing access requires sustained financial commitment. This creates a market gap where individuals who most need personalized guidance — beginners, those with health conditions, or those with dietary restrictions — are precisely those least likely to be able to afford it.

3. Target Users & Personas

The Fitness Beginner

An individual with limited workout experience seeking structured, safe entry into fitness. Needs progressive guidance, clear instructions, and confidence that the plan suits their current level. Highly susceptible to injury from generic plans designed for intermediate/advanced users.

The Health-Conscious Enthusiast

A regular exerciser looking to optimize their regimen based on a specific goal — fat loss, muscle gain, improved endurance, or general wellness. Has existing routines but wants evidence-based refinement tailored to their current status and constraints.

The Dietary-Restricted Individual

A person with vegan, vegetarian, or medically-mandated dietary requirements who has historically received nutrition plans that cannot be applied to their situation. Requires strict dietary compliance to be built into the plan's foundation, not treated as an afterthought.

The Health-Compromised User

An individual with pre-existing health conditions, injuries, or physical limitations that make standard plans inappropriate or dangerous. Needs intelligent avoidance of contraindicated movements and safety-first exercise selection.

4. Formal Problem Statement

Fitness enthusiasts, beginners, and health-conscious individuals cannot access truly personalized fitness and nutrition guidance that accounts for their unique combination of goals, fitness level, available equipment, dietary restrictions, and health conditions — without incurring the high cost and limited accessibility of professional personal trainers and nutritionists.

Generic online plans fail to address individual variation. Professional services are financially out of reach for most users. No existing automated solution integrates real-time evidence-based research with multi-dimensional personalization and scientific quality validation.

5. Root Cause Analysis

01

Information Generalization

Published fitness and nutrition content is deliberately created for broad audiences to maximize reach and commercial appeal. This systematically excludes edge cases — people with specific equipment constraints, health conditions, or dietary requirements — from receiving applicable guidance.

02

Lack of Multi-Dimensional Personalization

Even modern fitness apps typically personalize along only one or two dimensions (e.g., goal and fitness level). True personalization requires simultaneous consideration of at least six variables: goal, fitness level, equipment, dietary preferences, health conditions, and available training days — a complexity that cannot be encoded in simple rule-based systems.

03

Static Knowledge Problem

Most fitness content is based on knowledge that was current at the time of writing. Scientific understanding of exercise physiology and nutrition evolves continuously. Plans generated without access to current evidence-based guidelines risk being outdated, potentially recommending approaches that have since been refined or contradicted.

04

Absence of Quality Validation

Even AI-generated plans suffer from the risk of producing output that sounds authoritative but fails scientific scrutiny. Without an independent evaluation mechanism, users have no reliable way to assess whether a plan respects foundational principles such as progressive overload, muscle group balance, or safe recovery periods.

05

Cost and Accessibility Barriers

Professional-grade personalization requires either highly skilled human professionals or sufficiently sophisticated AI systems. Both are currently inaccessible to the majority of users — the former due to cost, the latter due to a lack of purpose-built solutions at this level of specificity.

6. Why an Agentic Approach?

Generating a truly personalized fitness and nutrition plan is not a single task — it is a collection of interdependent, specialized tasks that each require different forms of expertise. A single monolithic prompt cannot reliably perform research, structured planning, nutritional calculation, and scientific evaluation simultaneously. Attempting to do so produces plans that are superficially coherent but structurally weak in ways that may not be immediately apparent to the user.

Specialization	Each agent has a single, clearly scoped responsibility and a focused prompt, producing deeper, more reliable output per task than a generalist prompt would.

Sequential Grounding	Each agent builds on the verified output of the prior one. The Workout Planner cannot contradict the research findings because it receives them as direct input. The Nutrition Advisor is similarly constrained by established guidelines.

Independent Validation	The LLM-as-Judge agent evaluates the complete output against a 10-criterion scientific rubric without being biased by the generation process. This independence is only possible in a multi-agent structure.

Scalability	Each agent can be independently upgraded, replaced, or fine-tuned as needs evolve, without requiring a complete rebuild of the system.

7. Success Criteria

FORGE is considered to have solved the stated problem when the following conditions are met:

Success Criterion	Measurable Definition
Plans pass LLM-as-Judge evaluation	Generated plans consistently score ≥ 7/10 on the scientific rubric across diverse user profiles.
Full personalization coverage	All six input dimensions (goal, level, equipment, diet, health notes, training days) are demonstrably reflected in the output.
Dietary compliance enforced	Vegan/vegetarian plans contain zero non-compliant food items with appropriate macro adjustments.
Safety first for health conditions	Plans for users with noted health conditions actively avoid all contraindicated movement patterns.

Research grounded in current evidence	All guidelines surfaced by the researcher agent can be traced to real, current web sources retrieved at generation time.
Warnings surfaced for low-scoring plans	Any plan scoring < 6 displays prominent user-facing warnings, protecting the user from sub-standard output.

8. Constraints & Assumptions

8.1 Constraints

- The system does not provide medical advice and is not a substitute for qualified healthcare consultation.
- Plan generation is dependent on third-party APIs (Google Gemini, Tavily). Service interruptions in either will affect output quality.
- Output quality is bounded by the accuracy and availability of real-time web research at the time of generation.
- The LLM-as-Judge score is an AI-generated quality estimate, not a clinically validated safety assessment.
- The system generates plans for general wellness purposes only — not for competitive athlete preparation or clinical rehabilitation.

8.2 Assumptions

- Users provide honest and accurate information about their fitness level, health conditions, and goals.
- Real-time web search results via Tavily accurately represent current evidence-based guidelines.
- Users have access to a stable internet connection to use the deployed application.
- The Gemini 1.5 Flash model has sufficient capability to perform each specialized agent task at the required quality level.